Gradient Subspace Search: Compute-Efficient Post-Training via Low-Dimensional Weight Perturbation

A Follow-up to *Neural Thickets* (Gan & Isola, 2026)

Abstract

Gan & Isola (2026) show that large pretrained language models reside in *neural thickets* — dense regions of weight space where task-improving parameter vectors are abundant. Their algorithm, RandOpt, exploits this by randomly sampling N=5,000 weight perturbations, selecting the top-K, and ensembling predictions via majority vote. Despite matching or outperforming GRPO and PPO, RandOpt requires thousands of independent forward passes, which is prohibitive without a large parallel cluster. The paper also reports that fine-tuning loss landscapes exhibit *low-dimensional curvature*: a small number of directions dominate reward improvements. We ask whether this structure is captured by the task gradient — specifically, whether the top-K perturbations found by RandOpt concentrate in the gradient subspace. We first design a lightweight verification experiment to test this alignment directly. If confirmed, we propose **Subspace RandOpt**: restrict random search to the low-dimensional subspace defined by the top singular vectors of the task gradient, reducing N from 5,000 to ∼300–500 while maintaining competitive accuracy. The total cost is one backward pass (subspace construction) plus ∼500 forward passes — accessible on a single GPU.

Contents

1. Motivation and Problem Statement

Post-training of large language models (LLMs) typically relies on sequential, gradient-based optimization — PPO [Schulman et al., 2017], GRPO [Shao et al., 2024] — which requires expensive backpropagation and sequential updates. Gan & Isola (2026) challenge this assumption: in sufficiently large pretrained models, task-improving weight vectors are already dense around the pretrained initialization, making gradient-free random search viable.

Remaining bottleneck. Although RandOpt eliminates backpropagation, its compute cost is dominated by evaluating N perturbed models on a training set $\mathcal{D}_{\text{train}}$:

$$\text{Cost}_{\text{RandOpt}} = N \times C_{\text{fwd}},$$

where C_{fwd} is the cost of one forward pass over $\mathcal{D}_{\text{train}}$. With N=5,000 and models of 1.5B–8B parameters, this demands hundreds of GPUs running in parallel. Practitioners with limited hardware cannot replicate the reported results.

The underexploited finding. The paper notes that LLM fine-tuning landscapes have *low-dimensional curvature* [Liang et al., 2026]: only a small number of directions in weight space drive reward improvements. This is used as a post-hoc explanation but never exploited as a design principle. If top-K perturbations concentrate in the gradient subspace, searching in that subspace directly requires far fewer samples N to achieve good coverage.

Central research question.

> *Do the top-K weight perturbations found by* RandOpt *concentrate in the gradient subspace of the task reward — and if so, can searching directly in this subspace reduce N by $10\times$?*

2. Background

2.1 RandOpt (Gan & Isola, 2026)

Let $f_{\boldsymbol{\theta}} : \mathcal{X} \to \mathcal{Y}$ be a pretrained model with weights $\boldsymbol{\theta} \in \mathbb{R}^d$. RandOpt operates in two phases:

Training (random guessing). Sample N seeds $\{s_1, \ldots, s_N\}$ and noise scales $\{\sigma_i\}$:

$$\boldsymbol{\theta}'_i = \boldsymbol{\theta} + \sigma_i \cdot \boldsymbol{\varepsilon}(s_i), \quad \boldsymbol{\varepsilon}(s_i) \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d).$$

Evaluate each $\boldsymbol{\theta}'_i$ on $\mathcal{D}_{\text{train}}$ to obtain score v_i. Select top-K indices: $\mathcal{I}_{\text{top}} = \text{topK}(\{v_i\})$.

Inference (ensembling). For a test input x:

$$\hat{y} = \text{mode}\left(\left\{\arg\max_y f_{\boldsymbol{\theta}'_i}(y \mid x) \;\middle|\; i \in \mathcal{I}_{\text{top}}\right\}\right).$$

Key empirical findings relevant to this proposal:

- Solution density scales with model size: larger models have denser thickets, so fewer random draws are needed per high-quality solution.
- LLM fine-tuning landscapes exhibit *low-dimensional curvature* [Liang et al., 2026]: a small number of directions dominate reward improvements.

- Math reasoning tasks can be learned by updating as few as 13 parameters [Morris et al., 2026].
- Scaling N yields log-linear accuracy gains (Figure 7 of the paper), implying diminishing returns at large N.

2.2 Relevant Prior Work

Gradient subspace methods. Prior work on parameter-efficient fine-tuning shows that downstream adaptation is effectively low-dimensional [Aghajanyan et al., 2020]: gradient descent operates in a small subspace of the full parameter space. LoRA [Hu et al., 2022] exploits this by restricting weight updates to low-rank matrices. We ask whether the same subspace is relevant for gradient-free random search.

Fisher information and curvature. The Fisher information matrix (FIM) diagonal $F_{ii} = \mathbb{E}[(\partial \log p_\theta / \partial \theta_i)^2]$ measures parameter sensitivity. High-curvature directions are more likely to produce meaningful behavioral change under small perturbations. The empirical Fisher, computed from a small dataset, approximates the FIM cheaply via a single backward pass.

3. Research Plan

The project proceeds in two sequential phases. Phase 1 is a lightweight verification experiment that must be completed before Phase 2 begins. Phase 2 is only pursued if Phase 1 confirms the core assumption.

3.1 Phase 1: Verification — Does Gradient Subspace Alignment Hold?

3.1.1 The Core Assumption

Hypothesis 1 (Gradient Subspace Alignment). *The perturbation deltas $\Delta_i = \boldsymbol{\theta}'_i - \boldsymbol{\theta}$ of the top-K models selected by* RandOpt *have significantly higher projection energy onto the gradient subspace $\mathcal{S}_r$ than randomly selected perturbations of equal norm.*

3.1.2 Procedure

Algorithm 1 Subspace Alignment Verification

// Step 1: Run vanilla RandOpt (small N)
Sample N=500 perturbations: $\boldsymbol{\theta}'_i = \boldsymbol{\theta} + \sigma \cdot \boldsymbol{\varepsilon}(s_i)$
Evaluate on $\mathcal{D}_{\text{train}}$; select top-$K$=50 (set $\mathcal{I}_{\text{top}}$)
Collect deltas: $\Delta_i^+ = \boldsymbol{\theta}'_i - \boldsymbol{\theta}$ for $i \in \mathcal{I}_{\text{top}}$, Δ_j^- for K random non-top indices

// Step 2: Compute gradient subspace
$\mathbf{g} = \nabla_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}, \mathcal{D}_{\text{loc}})$ (1 backward pass on $|\mathcal{D}_{\text{loc}}|$=50 examples)
Reshape $\mathbf{g}$ into matrix $\mathbf{G}$; compute thin SVD: $\mathbf{U}, \mathbf{S}, \mathbf{V} = \text{SVD}(\mathbf{G})$
$\mathbf{P}_r = \mathbf{U}_{:,:r}$ for $r \in \{10, 50, 100, 200, 500\}$

// Step 3: Measure projection energy
for each delta $\Delta \in \{\Delta_i^+\} \cup \{\Delta_j^-\}$ **do**
    $\rho_r(\Delta) = \|\mathbf{P}_r \mathbf{P}_r^\top \Delta\|^2 \;/\; \|\Delta\|^2$
end for
Report $\bar{\rho}_r^+$, $\bar{\rho}_r^-$, and ratio $\bar{\rho}_r^+ / \bar{\rho}_r^-$ as a function of r

3.1.3 Additional Diagnostic Metrics

- **Correlation**: Does $\rho_r(\Delta_i)$ correlate with task score v_i across all N perturbations? If yes, subspace alignment predicts performance, strongly supporting the assumption.
- **Subspace stability**: Recompute $\mathbf{P}_r$ with 5 different random subsets of $\mathcal{D}_{\text{loc}}$. High variance indicates 50 examples is too small; increase $|\mathcal{D}_{\text{loc}}|$ or switch to Fisher.
- **Effective rank**: Minimum r to capture 50%, 80%, 95% of average top-K delta energy.

3.1.4 Interpretation and Decision Gate

Result	**Interpretation**	**Action**
$\bar{\rho}^+ \gg \bar{\rho}^-$ ($>2\times$)	Assumption holds	Proceed to Phase 2 (Subspace RandOpt)
$\bar{\rho}^+ \approx \bar{\rho}^-$	Gradient is the wrong basis	Test Fisher diagonal or PCA of top-K deltas as basis
Gap only for large r (>500)	Subspace is higher-dimensional	Test layerwise gradient subspaces
$\rho_r(\Delta_i)$ correlates with v_i	Alignment predicts task gain	Strong evidence; proceed and report correlation as a finding
$\mathbf{P}_r$ unstable across $\mathcal{D}_{\text{loc}}$ subsets	50 examples too small	Increase $|\mathcal{D}_{\text{loc}}|$; retry

3.1.5 Compute Budget for Verification

Step	Operation	Cost
Vanilla RandOpt (N=500)	500 forward passes on 200 examples	∼1–2 hrs
Gradient + SVD	1 backward pass on 50 examples + SVD	∼5 min
Projection analysis	Matrix multiplications only	∼1 min
Total		∼2 hrs, single GPU

3.2 Phase 2: Subspace RandOpt

3.2.1 Motivation

Standard RandOpt samples isotropically in $\mathbb{R}^d$ ($d \sim 7\times10^9$). Most perturbation energy falls in directions irrelevant to the task, requiring large N to find good solutions by chance. If Phase 1 confirms that top-K deltas concentrate in the gradient subspace $\mathcal{S}_r$ ($r \ll d$), we can restrict search to $\mathcal{S}_r$ directly, achieving equivalent coverage with far fewer samples.

3.2.2 Method

Step 1 — Subspace construction (one backward pass).

$$\mathbf{g} = \nabla_{\boldsymbol{\theta}}\mathcal{L}(\boldsymbol{\theta}, \mathcal{D}_{\text{loc}}), \qquad \mathbf{P}_r = \text{top-}r \text{ left singular vectors of } \mathbf{g} \text{ reshaped.}$$

Step 2 — Random search within the subspace. Replace full-space Gaussian noise with subspace-projected noise:

$$\boldsymbol{\theta}'_i = \boldsymbol{\theta} + \sigma \cdot \mathbf{P}_r\mathbf{z}_i, \qquad \mathbf{z}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_r).$$

$\mathbf{P}_r\mathbf{z}_i \in \mathbb{R}^d$ is a full-dimensional perturbation, but all energy is confined to $\mathcal{S}_r$. The effective search dimension drops from d to r.

Step 3 — Selection and ensembling. Identical to vanilla RandOpt: evaluate each $\boldsymbol{\theta}'_i$ on $\mathcal{D}_{\text{train}}$, select top-$K$, ensemble via majority vote.

3.2.3 Why Fewer Samples Suffice

To cover a fraction α of the relevant neighborhood, the number of samples required scales roughly with the search dimensionality. Reducing from $d \sim 7\times10^9$ to $r \sim 100$ means:

$$N_{\text{subspace}} \;\approx\; N_{\text{full}} \cdot (r/d)^{1/2} \;\ll\; N_{\text{full}}.$$

For r=100 and N_{full}=5000, this suggests $N_{\text{subspace}} \sim$ 300–500 is sufficient.

3.2.4 Full Algorithm

Algorithm 2 Subspace RandOpt

Require: Pretrained weights $\boldsymbol{\theta}$, localization set $\mathcal{D}_{\text{loc}}$, training set $\mathcal{D}_{\text{train}}$, subspace rank r, population N, ensemble size K, noise scale σ.

// **Subspace construction**
$\mathbf{g} \leftarrow \nabla_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}, \mathcal{D}_{\text{loc}})$
$\mathbf{U}, \mathbf{S}, \mathbf{V} \leftarrow \text{SVD}(\text{reshape}(\mathbf{g}))$
$\mathbf{P}_r \leftarrow \mathbf{U}_{:,\,:r}$

// **Random search in subspace**
for $i = 1, \dots, N$ **do**
    Sample $\mathbf{z}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_r)$
    $\boldsymbol{\theta}'_i \leftarrow \boldsymbol{\theta} + \sigma \cdot \mathbf{P}_r \mathbf{z}_i$
    $v_i \leftarrow \text{evaluate}(\boldsymbol{\theta}'_i,\, \mathcal{D}_{\text{train}})$
end for
$\mathcal{I}_{\text{top}} \leftarrow \text{topK}(\{v_i\})$

// **Inference**
$\hat{y} \leftarrow \text{mode}\Big(\{\arg\max_y f_{\boldsymbol{\theta}'_i}(y \mid x) \mid i \in \mathcal{I}_{\text{top}}\}\Big)$

3.2.5 Comparison to Vanilla RandOpt

Method	Search space	N	Backprop	Target acc. (GSM8K)
GRPO (200 steps)	—	—	200	∼82%
RandOpt (N=5000)	$\mathbb{R}^d$	5,000	0	∼87%
RandOpt (N=500)	$\mathbb{R}^d$	500	0	lower
Subspace RandOpt (ours)	$\mathcal{S}_r \subset \mathbb{R}^d$	**300–500**	**1**	**∼87%?**

3.3 Minor Extension: Residual-Guided Search

After Phase 2, if budget allows, we explore a simple complementary idea. Once a Round 1 ensemble $\mathcal{E}_1$ is built, identify the *residual* set $\mathcal{R} = \{j : \hat{y}_j^{(1)} \neq y_j\}$ of examples still answered incorrectly. A second round evaluates new perturbations *only on* $\mathcal{R}$:

$$v_i^{(2)} = \frac{1}{|\mathcal{R}|} \sum_{j \in \mathcal{R}} \mathbf{1}[f_{\boldsymbol{\theta}'_i}(x_j) = y_j].$$

This reduces evaluation cost per perturbation in Round 2 by a factor of $|\mathcal{R}|/|\mathcal{D}_{\text{train}}|$, and targets specialists for hard examples. Unlike AdaBoost, no reweighting or theoretical guarantees are claimed — this is stratified selection, not adaptive boosting. We include it as an ablation, not a core contribution.

4. Experimental Plan

4.1 Models and Benchmarks

Following the original paper, we use **Qwen2.5-1.5B-Instruct** and **Qwen2.5-3B-Instruct** as primary models, evaluated on tasks with verifiable outputs:

- **Mathematical reasoning**: GSM8K (primary), Countdown (secondary)
- **Code generation**: MBPP

Seq2seq and open-ended generation tasks are out of scope for this proposal, as majority-vote ensembling requires a discrete verifiable answer.

4.2 Baselines

1. RandOpt (N=5000, K=50) — full-compute upper bound
2. RandOpt (N=500, K=50) — compute-matched baseline (same N as Subspace RandOpt, isotropic noise)
3. GRPO (200 steps) — standard sequential gradient method

The critical comparison is **Subspace RandOpt (N=500) vs. RandOpt (N=500)**: same compute budget, different search geometry. Any accuracy improvement is attributable solely to the subspace structure.

4.3 Ablations

Subspace construction.

- Basis: gradient SVD vs. empirical Fisher diagonal vs. random projection (control)
- Rank $r \in \{10, 50, 100, 200, 500\}$
- Size of $\mathcal{D}_{\text{loc}} \in \{10, 50, 100, 200\}$

Generalization.

- Task transfer: compute subspace on GSM8K, run search on Countdown
- Model scale: repeat Phase 1 verification on Qwen2.5-3B to test whether alignment strengthens with scale (consistent with thicket scaling law)
- Diversity preservation: measure spectral discordance $\mathcal{D}$ [Gan & Isola, 2026] for subspace vs. full-space perturbations

4.4 Metrics

- Task accuracy (primary)
- Total FLOPs at equal compute budget
- Subspace energy ratio $\bar{\rho}_r^+$ (Phase 1 diagnostic, carried forward)
- Spectral discordance $\mathcal{D}$ (diversity preservation)

5. Hypotheses

Hypothesis 2. *Top-K perturbation deltas have subspace energy ratio $\bar{\rho}_r^+ \geq 2\bar{\rho}_r^-$ for $r \leq 200$, confirming gradient subspace alignment.*

Hypothesis 3. *Subspace RandOpt with N=500 achieves accuracy within 2% of vanilla* RandOpt *with N=5000, and outperforms* RandOpt *with N=500 (isotropic), on GSM8K.*

Hypothesis 4. *The subspace $\mathbf{P}_r$ computed on $\mathcal{D}_{\text{loc}}$ (50 examples) is stable across different random subsets, indicating that the gradient direction is a consistent property of the task, not an artifact of example selection.*

Hypothesis 5. *Spectral discordance $\mathcal{D}$ of subspace perturbations is not significantly lower than for full-space perturbations, preserving the specialist diversity that makes ensembling effective.*

6. Risks and Mitigations

Risk	**Mitigation**
Assumption fails ($\bar{\rho}^+ \approx \bar{\rho}^-$)	Switch basis to empirical Fisher or PCA of top-K deltas; report null result as a finding about the geometry of RandOpt.
Subspace search reduces specialist diversity	Measure $\mathcal{D}$ vs. r; report Pareto frontier between diversity and compute savings.
Gradient unstable at small $\lvert\mathcal{D}_{\text{loc}}\rvert$	Increase $\lvert\mathcal{D}_{\text{loc}}\rvert$; ablate stability explicitly (Section 4.3).
Performance gap vs. full RandOpt is large	Compare at equal FLOPs, not equal N; one backward pass buys substantial compute savings even if accuracy gap is nonzero.

7. Timeline

Phase		Tasks	Duration
0	Reproduce	Vanilla RandOpt on GSM8K, Qwen2.5-1.5B	1 week
1	Verify	Subspace alignment experiment; report $\bar{\rho}^+$ vs. $\bar{\rho}^-$	2 weeks
2	Method	Subspace RandOpt implementation + main results	3 weeks
3	Ablations	Basis, rank, $\lvert\mathcal{D}_{\text{loc}}\rvert$, transfer, diversity	3 weeks
4	Write-up	Paper draft	3 weeks
Total			12 weeks

Decision gate: If Phase 1 shows $\bar{\rho}^+ \approx \bar{\rho}^-$, Phases 2–3 pivot to characterizing *what basis* does explain top-K delta geometry (Fisher, Hessian eigenvectors, or PCA of deltas from related tasks). This pivot is itself a publishable finding.

8. Summary of Contributions

1. **Empirical finding (Phase 1)**: The first direct measurement of whether RandOpt's top-K perturbations align with the task gradient subspace — a question the original paper raises but does not answer.
2. **Subspace RandOpt (Phase 2)**: A method that restricts gradient-free random search to the low-dimensional gradient subspace, achieving comparable accuracy to vanilla RandOpt with 10× fewer perturbation evaluations and a single backward pass.
3. **Practical impact**: Effective post-training on a single GPU, removing the cluster-scale compute requirement of the original method.

9. Summary of Experiments

Experimental Setup. Given a LLM f_θ, training set $D_{train} = \{(x_0, y_0), (x_1, y_1), ..., (x_m, y_m)\}$, and test set $D_{test} = \{(x_0, y_0), (x_1, y_1), ..., (x_n, y_n)\}$, we fist use SFT to cacluate the graident ∇f_θ. Then the importance of each parameter M_r is obtained. With a hyperparameter ratio k, the most important Top-K parameters are selected with a binary mask M. Following Neural TicketsGan & Isola [2026], we inject random noise into the Top-K parameters in θ according to M. Besides, we apply greedy updates after every 50 samples.
Observation. Our method has much more overfitting than the baseline.

Table 1: (200)Training Performance (Best Checkpoint at Each Stage)

	Baseline		Sparsity		Sparsity (No Update)	
Checkpoint	Reward	Acc	Reward	Acc	Reward	Acc
100	0.1878	15.00%	0.1820	**15.50%**	0.1492	12.00%
200	0.1878	15.00%	**0.2462**	**21.00%**	0.1492	12.00%
300	0.1878	15.00%	**0.2813**	**25.00%**	0.1492	12.00%
400	0.1878	15.00%	**0.2813**	**25.00%**	0.1492	12.00%
500	0.1878	15.00%	**0.2813**	**25.00%**	0.1492	12.00%
600	0.1878	15.00%	**0.2813**	**25.00%**	0.1555	12.00%
700	**0.1881**	14.00%	**0.2813**	**25.00%**	0.1555	12.00%

Figure 1: Caption

Table 2: Test Performance (Majority Voting Accuracy)

Checkpoint	**K**	**Baseline**	**Sparsity**	**Sparsity (No Update)**
100	1	12.50%	8.35%	9.40%
	5	26.75%	20.60%	21.75%
	10	33.15%	29.20%	28.30%
	20	39.45%	34.55%	36.85%
	50	45.35%	42.55%	43.10%
200	1	12.50%	8.00%	9.40%
	5	29.30%	23.05%	23.20%
	10	35.50%	30.40%	31.00%
	20	**40.90%**	36.50%	37.05%
	50	46.80%	42.55%	44.75%
300	1	12.50%	7.95%	9.40%
	5	**31.20%**	19.90%	24.00%
	10	**36.35%**	26.40%	31.35%
	20	40.85%	34.15%	37.95%
	50	**46.90%**	40.45%	45.55%
400	1	12.50%	7.95%	9.40%
	5	30.00%	20.55%	24.75%
	10	35.25%	29.25%	32.70%
	20	40.30%	35.95%	38.65%
	50	45.65%	41.20%	45.95%
500	1	12.50%	7.95%	9.40%
	5	30.00%	20.55%	24.75%
	10	35.60%	29.25%	32.70%
	20	40.80%	35.55%	38.60%
	50	46.45%	41.60%	45.50%
600	1	12.50%	7.95%	9.85%
	5	28.35%	20.55%	25.45%
	10	35.25%	29.40%	33.65%
	20	40.30%	34.65%	39.65%
	50	**47.00%**	41.65%	45.65%
700	1	**15.95%**	7.95%	9.85%
	5	29.70%	20.55%	25.80%
	10	34.10%	29.40%	33.60%
	20	40.70%	34.80%	40.40%
	50	46.25%	41.70%	46.50%

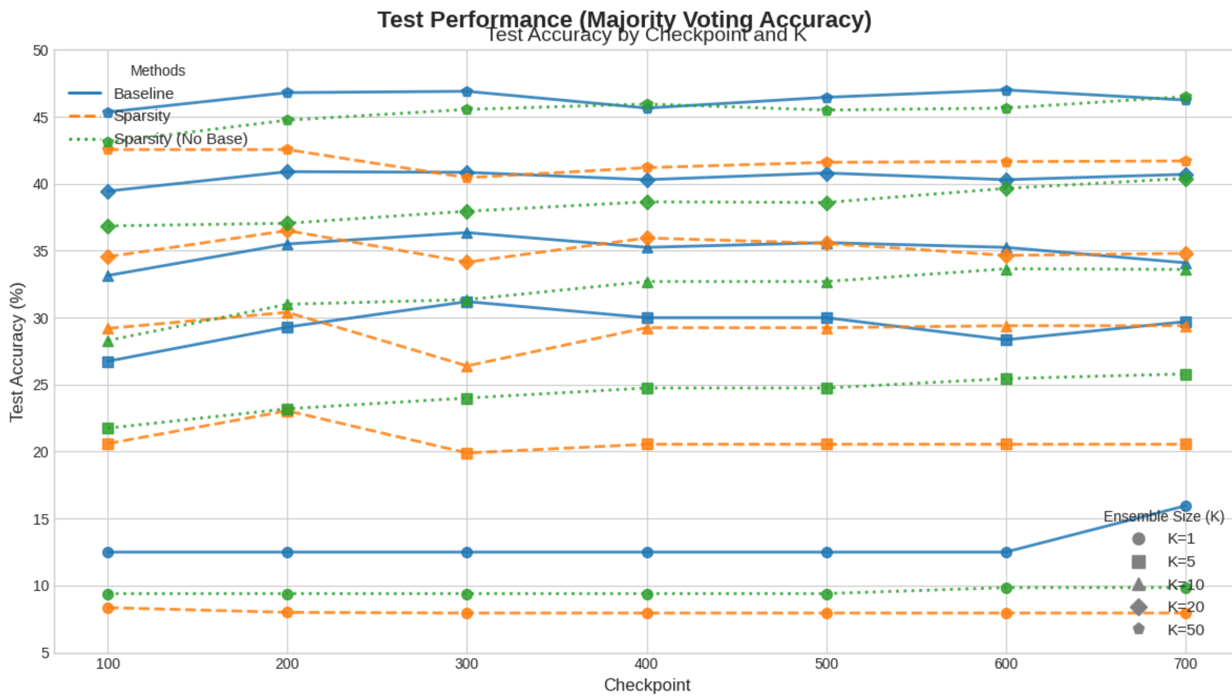

Figure 2: Caption

Table 3: (50)Training Accuracy Comparison (Best Top-1 Acc per Checkpoint)

Checkpoint	**100**	**200**	**300**	**400**	**500**	**600**	**700**	**800**	**900**	**1000**	**1100**
Baseline	14.00%	22.00%	22.00%	22.00%	26.00%	26.00%	26.00%	26.00%	26.00%	26.00%	26.00%
Ours	18.00%	24.00%	24.00%	28.00%	32.00%	32.00%	32.00%	32.00%	32.00%	34.00%	34.00%

Checkpoint	**1200**	**1300**	**1400**	**1500**	**1600**	**1700**	**1800**	**1900**	**2000**
Baseline	26.00%	26.00%	26.00%	26.00%	26.00%	26.00%	26.00%	26.00%	26.00%
Ours	34.00%	34.00%	34.00%	34.00%	34.00%	34.00%	34.00%	36.00%	36.00%

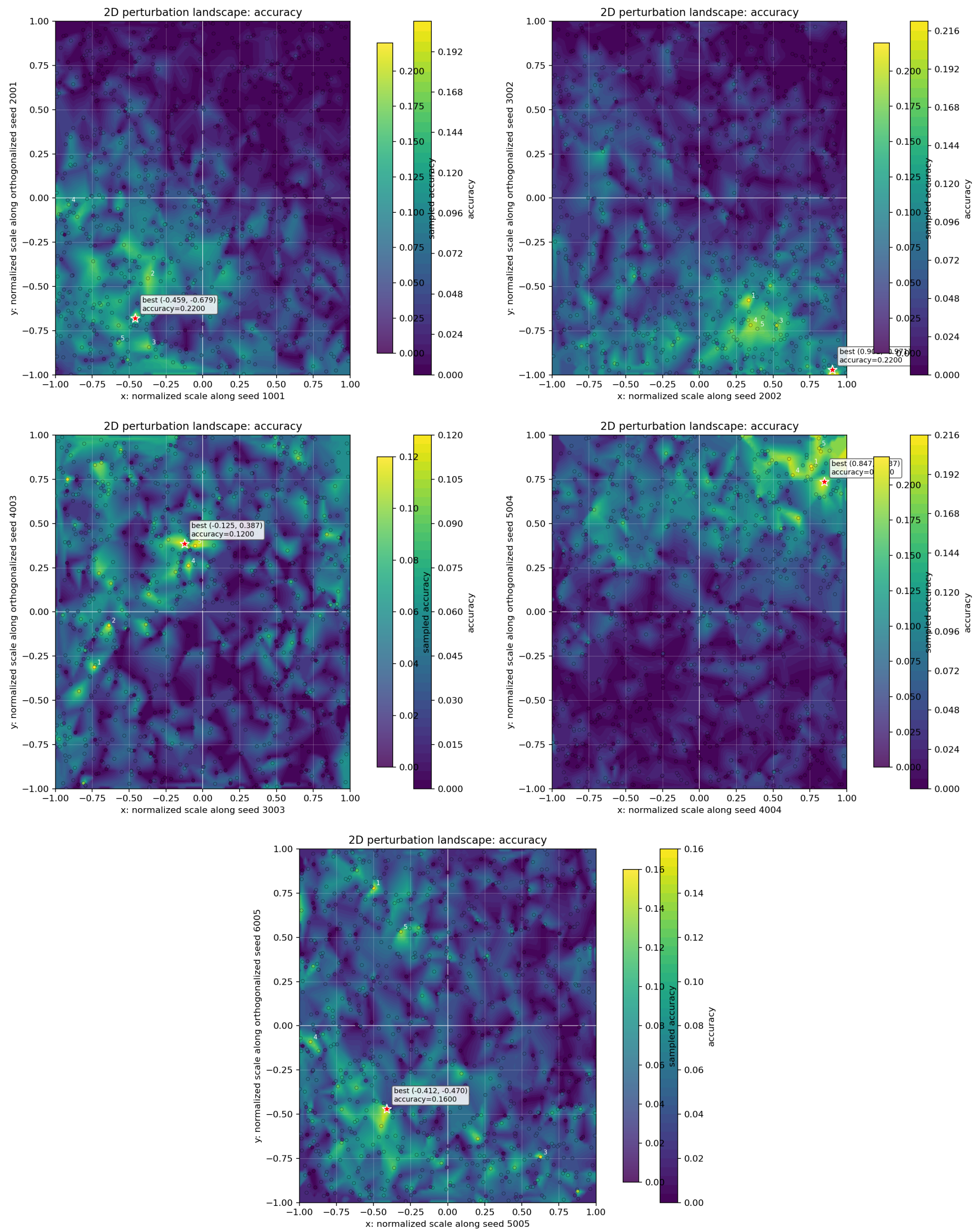

Figure 3: Injecting noise with two directions.

Table 4: (150)Test Accuracy Comparison across Different Checkpoints and Top-K Candidates

	K=1		**K=5**		**K=10**		**K=20**		**K=50**	
Checkpoint	**Base**	**Ours**	**Base**	**Ours**	**Base**	**Ours**	**Base**	**Ours**	**Base**	**Ours**
100	13.91%	7.28%	23.84%	22.52%	29.14%	29.80%	37.75%	41.72%	43.05%	51.66%
200	9.27%	6.62%	23.84%	21.85%	30.46%	32.45%	39.74%	36.42%	45.70%	47.68%
300	9.27%	6.62%	25.17%	23.84%	31.13%	31.79%	39.74%	38.41%	47.68%	45.70%
400	9.27%	6.62%	22.52%	24.50%	29.14%	31.13%	37.09%	43.05%	47.02%	50.33%
500	12.58%	8.61%	27.81%	17.22%	34.44%	28.48%	39.74%	36.42%	50.99%	47.68%
600	12.58%	8.61%	29.14%	17.22%	35.76%	28.48%	41.72%	35.76%	49.01%	49.01%
700	12.58%	8.61%	27.81%	17.22%	36.42%	28.48%	40.40%	37.09%	47.68%	47.02%
800	12.58%	8.61%	27.81%	17.22%	35.76%	28.48%	42.38%	36.42%	50.33%	45.70%
900	12.58%	8.61%	27.81%	17.22%	33.11%	25.83%	39.74%	35.10%	50.99%	45.03%
1000	12.58%	8.61%	27.81%	20.53%	33.11%	25.83%	38.41%	36.42%	47.68%	43.71%
1100	12.58%	8.61%	27.81%	20.53%	33.11%	28.48%	40.40%	35.76%	48.34%	44.37%
1200	12.58%	8.61%	27.81%	21.85%	33.11%	30.46%	39.07%	35.76%	47.68%	43.05%
1300	12.58%	8.61%	29.14%	21.85%	31.79%	29.80%	37.75%	37.75%	47.02%	44.37%
1400	12.58%	8.61%	29.14%	19.87%	35.76%	27.81%	41.06%	36.42%	45.70%	43.71%
1500	12.58%	8.61%	29.14%	19.87%	35.76%	26.49%	40.40%	35.10%	45.70%	44.37%
1600	12.58%	8.61%	28.48%	21.85%	33.77%	27.81%	41.06%	35.10%	45.03%	41.72%
1700	12.58%	8.61%	27.81%	21.19%	33.11%	27.81%	40.40%	34.44%	45.03%	41.72%
1800	12.58%	8.61%	27.81%	23.18%	33.11%	27.81%	40.40%	35.10%	45.70%	42.38%
1900	12.58%	9.27%	27.81%	22.52%	33.11%	27.81%	40.40%	36.42%	47.02%	43.05%
2000	12.58%	9.27%	27.81%	24.50%	35.10%	28.48%	40.40%	33.11%	47.68%	41.72%

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